Semantic Retrieval for Agentic AI

Ontology-defined meaning + deterministic execution across federated data

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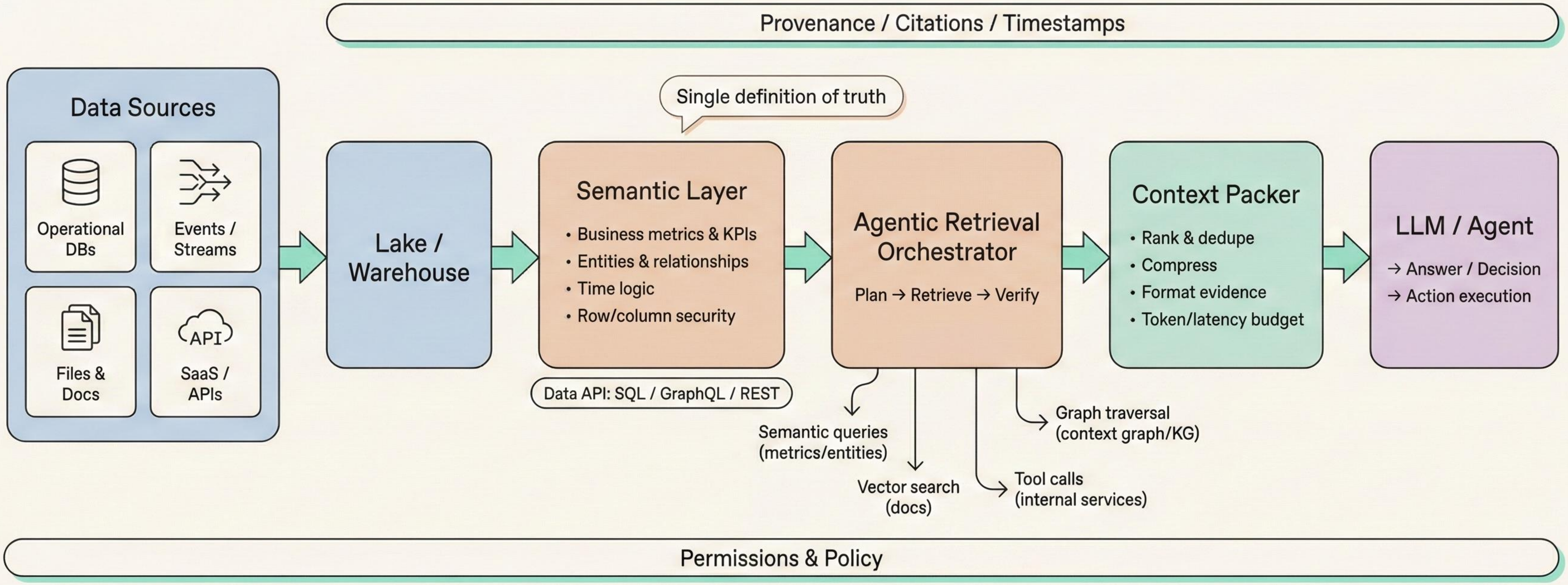
The Promise

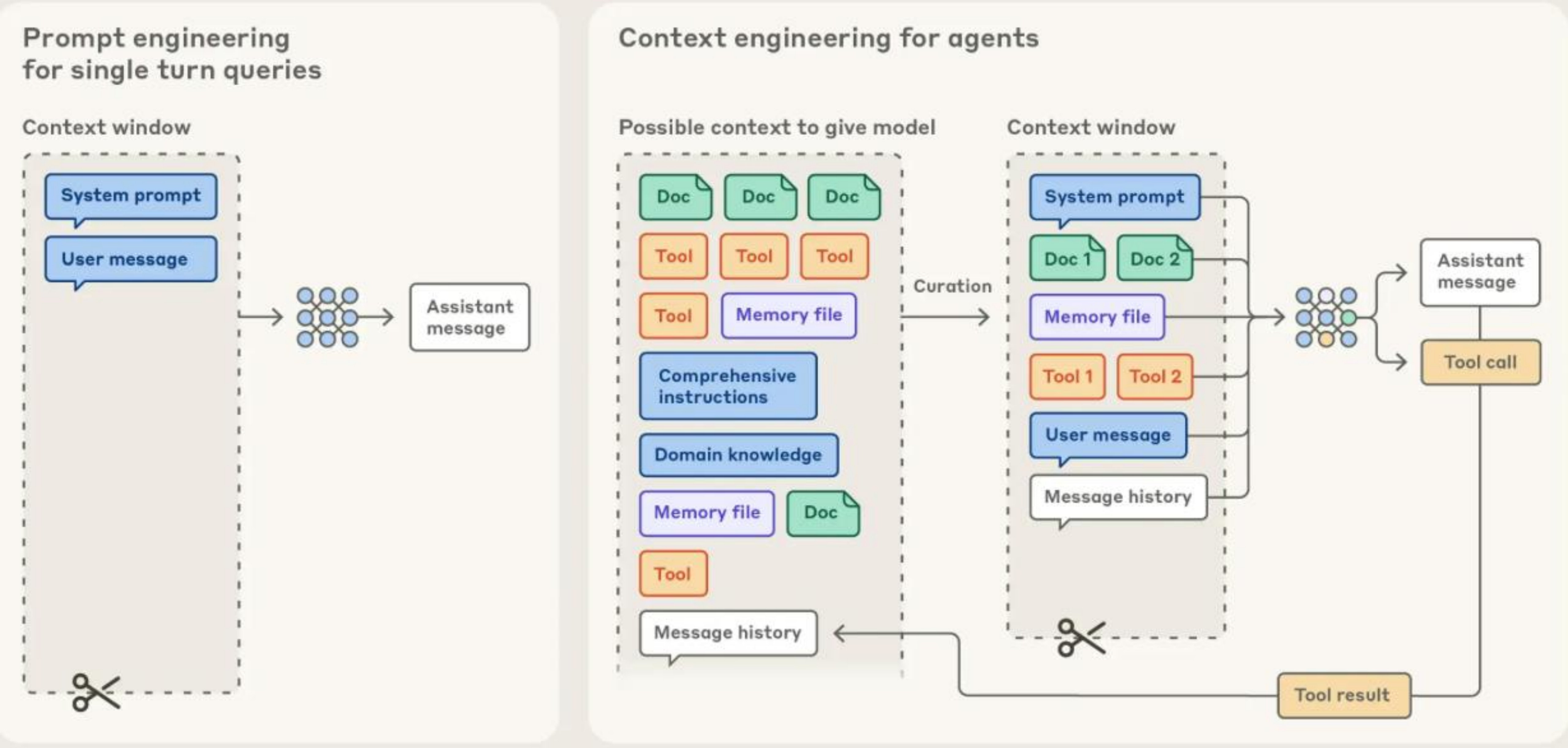
“"Just connect all the data sources and let the AI answer questions."”

- Data is federated; meaning is fragmented.
- RAG adds text context, but not KPI truth, entity identity, or policy.
- Agents need a governed semantic surface they can plan and compile against.

Core Stack: Data → Semantic Layer → Agentic Retrieval → LLM



Prompt engineering vs. context engineering



Federated agentic retrieval: where it breaks

Pipeline failure modes in a federated data space

User question | Intent router | Planner & tools | Structured query | Unstructured retrieval | Fusion + verify

- Ambiguous terms
- Time / Temporal intent mismatch
- Wrong route
- Non-determinism
- Routing ambiguity (same table, 2 stores)
- Metric drift / definition inconsistency
- Join / grain
- Relevance
- No citations / low explanability
- Merge w/o proofs
- Conflicts
- Query-language mismatch (SQL/KQL/T-SQL/Cosmos)
- Entity Identity fragmentation (IATA≠ICAO)
- Silent failures (plausible wrong answers)
- Freshness mismatch (stale vs live)
- Unsafe tools / injection risk
- Structured / unstructured merge – top5 airports by delay rate and what do security narratives say about them
- Cross-source joins without foreign keys – Which flights were affected by active SIGMETs?

Our Federated Data Space

8 sources across 5 storage technologies (and 5 query languages)

PostgreSQL

ASRS incidents
Airport/runway refs
Ops overlay tables

Fabric Eventhouse

KQL streams
OpenSky live flights
SIGMET / G-AIRMET

AI Search (Safety)

Vector + semantic
ASRS narratives
~240K chunks

AI Search (Reg + Ops)

Regulations + airport docs
2 indexes
~2,055 docs

Cosmos DB

Active NOTAMs
Partitioned by /icao
25 docs / 8 airports

Knowledge Graph

Dependency edges
flight→crew→MEL→turnaround
500K+ edges

Fabric SQL Warehouse

BTS on-time reporting
Airline delay performance,
Delay causes breakdown
~500K rows

One question

Needs: SQL + KQL + Vector + NoSQL
...and the joins between them

Semantic Model v2 = a stack

Three layers that jointly make agents reliable.

Business semantics
(Metrics, dimensions, time, security)

Entity semantics - Ontology + graph (Concepts, relations, constraints)

Retrieval semantics
(Vector/Hybrid indexes + citations)

Microsoft mapping

- **Power BI semantic model**
- **Fabric IQ ontology + graph**
- **Foundry IQ KB (AI Search indexes)**

Key idea:

Agents query semantic surfaces,
not tables or blobs.

Microsoft Semantic Intelligence Stack

Fabric IQ + Data Agents + Foundry IQ form the backbone

Microsoft provides

Fabric IQ

Ontology + graph semantics
Disambiguation, entity IDs, constraints

Fabric Data Agents

Deterministic NLQ → SQL/KQL/DAX
Runs over governed sources

Foundry IQ + Azure AI Search

Knowledge bases + agentic retrieval
Hybrid retrieval + citations

Orchestration + verification

Query routing across structured + unstructured
Verification checks + conflict handling
Typed tool outputs + traceability

Cross-cutting

Entra/Fabric security
Observability (App Insights / Monitor)
Caching + budgets + SLOs
Eval harness for correctness

This is how "meaning" becomes a reusable, governed tool surface for agents.

How Fabric IQ uses the ontology (routing + retrieval)

NL → ontology query → routed execution → typed results → merged answer.

1) User asks a question

- “Which delayed flights were affected by active SIGMETs near destination?”

2) NL2Ontology (semantic parsing)

- Targets entity types + relationships
- Produces a structured ontology query

3) Bindings route to the best engine

- Flights → SQL/DAX
- SIGMET stream → KQL
- Impact traversal → GQL (Graph)

Typed outputs

- Flights (ids + delay mins)
- Hazards (time + geometry)
- Links (AFFECTS/OVERLAP)

Merge + provenance

- Explain join path + time window
- Optionally attach citations from Foundry IQ (docs)

Microsoft mapping (minimal)

Fabric IQ

- Ontology + bindings + graph semantics

Fabric Data Agent

- NL interface over governed sources

Foundry IQ / AI Search

- Unstructured KB + citations (optional)

Thank you...

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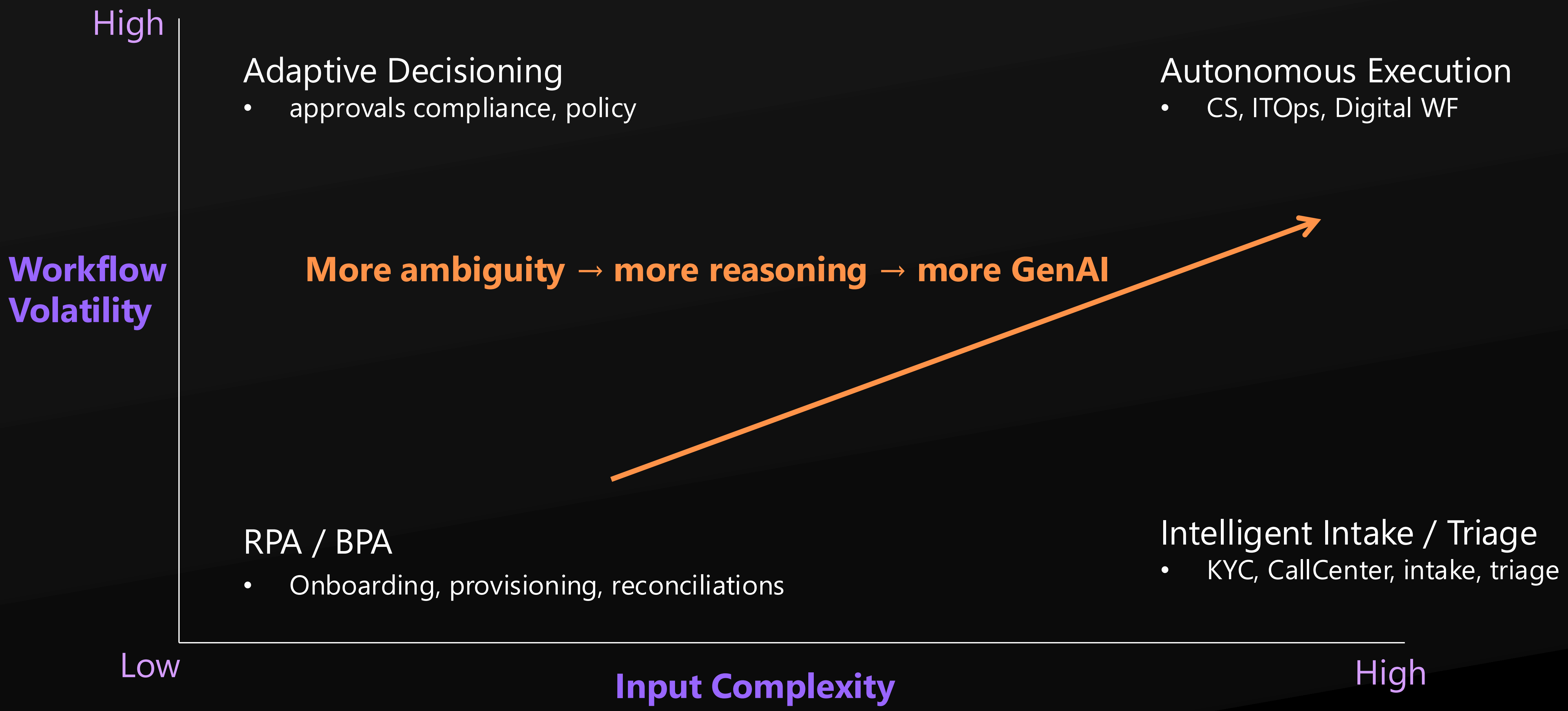
From Rules to Reasoning – AI Automation with GenAI



When Reasoning Helps Beyond Classification

Cross Domain Context Integration

Fuses Transactional, Behavioral and Conversational data fusion

Constraint Driven Optimization

Multi-constraint optimization e.g. personal & derived preferences, business, risk and compliance rules

Temporal Reasoning

Understands sequence, timing, and cause–effect in user journeys and generates what-if analysis

Conversational Explanation & Empathy

Explains actions, charges, and policies in natural language

Privacy-preserving segmentation & clustering

Demo storyboard: one question, three channels

Show how the semantic stack makes answers predictable + explainable.

1. Ask an ambiguous business question
2. Structured: KPI delta via Fabric Data Agent
3. Graph: resolve impacted entities via Fabric IQ
4. Unstructured: cite incident timeline via Foundry IQ
5. Synthesize: answer + provenance + “why”

Wow moment

Edit a semantic rule → re-run